

CSE-321 (Fall 2015)
HW5
Submission Deadline: December 23th - 17:00

Problem: IDO wants to build some piers for sea buses from Besiktas through the north of Istanbul. After some investigation, n candidate locations are defined with their distance to Besiktas and the expected annual profit from that pier. The problem is that IDO wants to leave at least K meters distance from one pier to the next while maximizing the total profit.

Design a dynamic programming algorithm to find a subset of candidate piers which allows maximum profit. In your report you need to write your algorithm as a pseudocode and make a formal computational complexity analysis of your algorithm (in terms of big-Oh etc. notation).

Subsequently, you need to write a (small) Java program that implements your algorithm.

The method that you need to implement in your program takes as input two arrays and return the indices of selected piers as shown below:

```
public static ArrayList<Integer> findBestLocationSubset(int [ ] profits, int [ ] distances)
```

In addition to correctness, you also need to make a performance evaluation of your algorithm as follows:

Generate 100 random distance and profit arrays for varying number of pier candidates and plot the results of your algorithm as a function of the number of visited cells in the matrix you use. Each point in your plot should correspond to the sample mean of 100 random arrays generated according to that particular setting.

Show your results as an Excel file attached to your report. The figure should be in your report, whereas the data should be as a separate Excel file.

In your report, make a discussion about the possible real-life applications of this problem. Your report and program are equally important and will be evaluated separately.

*Upload your source files and documentation to moodle and bring your report to room 118.

